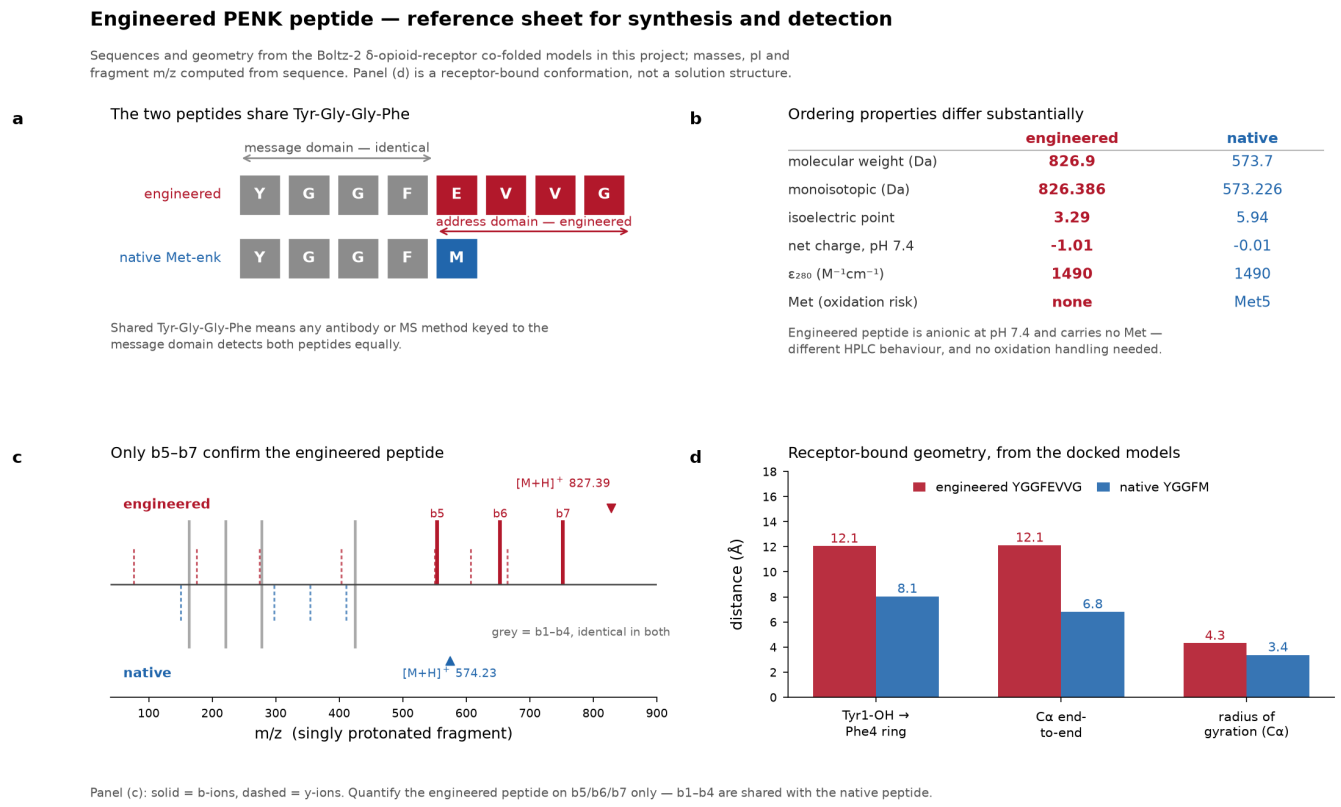
Engineered PENK peptide — wet-lab reference sheet

Peptide: engineered enkephalin analog **YGGFEVVG** (the octapeptide the therapy plasmid is designed to release, ×7 per proprotein). **Comparator:** native Met-enkephalin **YGGFM**. **Date:** 2026-08-06. **Status:** design reference — peptide not yet synthesised or assayed in-house.

0. What this sheet is for

Three separate uses, and they need different things from it:

1. **Ordering synthetic standards** — §1 gives the sequences and the physicochemical properties a peptide vendor and your HPLC need.
2. **Detecting the peptide in transfected-cell supernatant** — §2 gives the MS transitions, and states which ones are useless.
3. **Structural figures / MD setup** — §3 covers the PDB files, and §4 states the limit on what they can be used for.



Reference sheet at a glance. (a) The two peptides share the Tyr-Gly-Gly-Phe message domain — grey — so any method keyed to it detects both. (b) Properties for ordering and HPLC. (c) MS/MS fragments: b1–b4 coincide exactly, only b5–b7 are specific to the engineered peptide. (d) Receptor-bound geometry from the Boltz-2 δ-opioid-receptor models.

1. Synthetic standards to order

	Engineered	Native (comparator)
Sequence (N→C)	H-Tyr-Gly-Gly-Phe-Glu-Val-Val-Gly-OH	H-Tyr-Gly-Gly-Phe-Met-OH
One-letter	`YGGFEVVG`	`YGGFM`
Length	8	5
Average MW	**826.90 Da**	573.66 Da
Monoisotopic MW	**826.3861 Da**	573.2257 Da

	Engineered	Native (comparator)
Isoelectric point	**3.29**	5.94
Net charge at pH 7.4	**−1.01**	−0.01
ε_{280}	1490 $M^{-1}cm^{-1}$ (1 Tyr)	1490 $M^{-1}cm^{-1}$ (1 Tyr)
Termini	free amine / **free acid**	free amine / free acid
Stereochemistry	**all-L**	all-L

Order both. The native peptide is not optional — it is the comparator every δ-vs-μ selectivity claim in this project is made against, and it is also the species already present in the cells you transfect.

Three ordering details that are easy to get wrong

- **Free acid C-terminus, not amide.** The proprotein releases this peptide by proteolysis at a downstream dibasic site, which leaves a free carboxylate. Vendors default to C-terminal amide on many peptide syntheses; an amide would be a different molecule (−0.98 Da, and one fewer negative charge). Specify it.
- **All-L.** No D-amino acids anywhere. (Deltorphin-type analogs in the separate cycle-3b work do carry a D-residue at position 2 — that is a different peptide, do not conflate the orders.)
- **No Met in the engineered peptide.** This is an advantage: the native pentapeptide's Met5 oxidises to the sulfoxide (+16 Da) on storage and during handling, and you will see it as a shoulder in the native standard's LC-MS. The engineered peptide has no oxidisable residue.

Handling

Both are short, unstructured, and have no Cys — so no disulfide handling and no refolding. Reconstitute in water or dilute aqueous buffer; the engineered peptide is anionic at neutral pH (pI 3.29) so it stays soluble, but at pH ~3.3 it will be at its solubility minimum — avoid formic-acid-heavy buffers for stock storage (fine for the LC mobile phase). Aliquot and freeze; avoid repeated freeze-thaw.

Quantify by A_{280} using ε = 1490 $M^{-1}cm^{-1}$ — **identical for both peptides**, since both have exactly one Tyr and no Trp. Convenient: the same extinction coefficient, different masses, so A_{280} gives molar concentration directly and comparably.

---

2. Detecting the peptide by LC-MS/MS

Precursors

Peptide	$[M+H]^+$	$[M+2H]^{2+}$
Engineered YGGFEVVG	**827.393**	414.200
Native YGGFM	**574.233**	287.620

253.16 Da apart — trivially resolved at the precursor level.

The critical constraint: b1–b4 are identical in both peptides

Both peptides begin `YGGF`. The b-ion series is therefore **numerically identical** for the first four fragments:

Fragment	Engineered	Native	Discriminating?
b1	164.071	164.071	**no**
b2	221.092	221.092	**no**
b3	278.114	278.114	**no**
b4	425.182	425.182	**no**
b5	**554.225**	—	yes
b6	**653.293**	—	yes
b7	**752.362**	—	yes

Quantify on b5, b6, or b7 only. A method built on b2/b3/b4 — which are the strong, commonly-chosen fragments for enkephalins in the literature — cannot distinguish the engineered peptide from the native background, and you will be measuring the sum of both. This is the single most likely way to get a false positive from this assay.

y-ions: no coincidences between the two peptides at all, so any y-ion is discriminating. Engineered y-series: y7 664.330, y6 607.309, y5 550.287, y4 403.219, y3 274.176, y2 175.108, y1 76.039.

(All fragment m/z are singly-protonated monoisotopic, computed from sequence. No modifications assumed.)

Antibody detection — a warning

The same logic applies, and worse. Commercial anti-Met-enkephalin antibodies are raised against the `YGGF` message domain, because that is the conserved opioid motif. Such an antibody **will detect both peptides** and cannot be used to show that the engineered peptide was produced. If you need immunodetection, it has to be against the `EVVG` C-terminal extension, which means a custom antibody — and a 4-residue epitope is a hard ask. **MS is the correct method here, not ELISA.**

---

3. Structure files

Two PDB files accompany this sheet:

File	Contents
`engPENK_octapeptide_YGGFEVVG_DORbound.pdb`	engineered octapeptide, 8 residues, 58 atoms
`native_MetEnkephalin_YGGFM_DORbound.pdb`	native pentapeptide, 5 residues, 39 atoms

Both extracted from the Boltz-2 co-folded complexes with the human δ-opioid receptor already generated in this project (chain B of each), renumbered 1..N, chain A. Backbone verified contiguous (Cα–Cα 3.77–3.82 Å, no breaks). These are the exact coordinates the reported binding energies were computed on.

Receptor-bound geometry:

Metric	Engineered	Native
Tyr1-OH → Phe4 ring centroid	12.05 Å	8.05 Å
Cα end-to-end	12.12 Å	6.81 Å
Radius of gyration (Cα)	4.31 Å	3.38 Å

The Tyr1–Phe4 spacing is the **message-domain pharmacophore distance** — the geometry that determines opioid-receptor engagement. It differs by 4.0 Å between the two bound poses, i.e. the engineered peptide adopts a substantially more extended conformation in the same binding site. That is a model prediction, and it is the structural correlate of the predicted selectivity shift.

Model confidence (parent complexes): ipTM 0.972 (engineered) / 0.979 (native); peptide-chain pTM 0.981 / 0.983; interface pair ipTM 0.809 / 0.757.

---

4. Limits — read before using the structures

A linear octapeptide has no structure in isolation. Enkephalins are canonically disordered ligands; the coordinates above are receptor-bound conformations, one member of a broad solution ensemble. They are valid for rendering the bound pose, measuring contacts, and as MD starting coordinates **inside the receptor**. They are not "the structure of the engineered enkephalin", not comparable to a solution NMR ensemble, and not a rigid body for re-docking.

The selectivity prediction is computational and untested. The project's MM-GBSA gave ΔΔG_sel ≈ +4.05 kcal/mol favouring δ over μ for the engineered peptide. That is a prediction from a docked model, not a measurement. The bench test is a δ/μ binding or functional assay against both synthetic peptides — see §5.

No μ-receptor structures exist in the artifact store. Only the δ complexes were saved. A structural δ-vs-μ comparison would need those models regenerated.

5. What the bench should measure

The modelling makes exactly two falsifiable claims about this peptide. Both are cheap to test once the standards are in hand:

1. **The peptide is produced and secreted.** Transfect, collect supernatant, LC-MS/MS on the b5/b6/b7 transitions. Accept: engineered peptide detectable above the untransfected control. This tests the cassette, the signal peptide, and the processing sites all at once.
2. **It prefers δ over μ.** Radioligand competition or functional (cAMP / β-arrestin) assay at both receptors, engineered vs native peptide, same day, same cells. Accept: δ/μ ratio higher for engineered than native. If it is not, the central pharmacological hypothesis is wrong — that is a real result and should be reported as one, not re-run until it agrees.

Note that claim 1 can succeed while claim 2 fails, and they are worth separating in the experimental record for exactly that reason.

6. Provenance

Sequences read from the coordinate files, not transcribed. Masses, pI, net charge, ε_{280} and all fragment m/z computed from sequence (average and monoisotopic residue masses; Henderson–Hasselbalch with standard side-chain pKa values). Geometry measured directly from the PDB files. Every number in this sheet was verified programmatically against the computation before release. Design intent and the ΔΔG values come from the project's computational audit.